The `circledsteps` package: circled numbers, circled steps and more

Romano Giannetti <romano.giannetti@gmail.com>

June 16, 2021

1 Introduction

This package was born thanks to the discussion on tex.stackexchange.com, and with the help of the several authors that contributed to the answers. The base idea is to have a command that can create arbitrary "circled" text (numbers or letters) and that can work in normal text **and** in a node in a `tikzpicure` or derivative. That forbid to use `TikZ` itself for the circles — you can't safely nest `tikzpicture`s.

This package provides two things: the first one, macros to generate the circled text that use the original `pict2e` method, and then a (simple, to be taken as an example) set of macro to generate sequential circled numbers that can be referenced afterward.

The package loads (if not already loaded) the packages `pict2e`, `pictures`, `xcolor` and `pgfkeys` (the latter one is automatically loaded by `tikz`; you do not need `tikz` to use just `circledsteps`). Moreover, `etoolbox` is loaded to modify `pict2e` to have filled ovals.

The manual is for version v1.3.1. The main changes between version are listed below.

v1.3 Minor fixes to the manual; added a warning in case the patching of `picture.sty` fails.

v1.2 Added optional argument to the commands and make them sensitive to math mode (thanks to `@tallmarmot` for the suggestion and example code); fixed several bugs.

Added `fill color` for the circles background.

v1.1 Default to normal colors; keep the "red/blue" theme as optional. If you use the package option `redblue` or `legacy` you will have the same effect than

```
1  \pgfkeys{/csteps/inner color=red, /csteps/outer color=blue}
```

v1.0 Initial release.

2 Basic commands

The basic commands are:

`\Circled[]{}`	circled text using the package colors, text on the baseline
`\CircledTop[]{}`	circled text using the package colors, circle on the baseline
`\CircledText[]{}`	circled text using the current color[1]

[1] it may fail when used in headers or footers, in the sense it can select the wrong color in case of page breaks.

The parameters for the output are controlled using `pgfkeys`; you can change them with `\pgfkeys` or `\tikzset`; you can obviously limit the effect of change using normal LaTeX scoping rules. Also, you can specify parameters (without the initial `/csteps/` as optional arguments to the commands.

The pseudo-color `none` means that the current color is used for both the text and the circle.

Parameter	Meaning	Default value
`/csteps/inner ysep`	vertical spacing	4pt
`/csteps/inner xsep`	horizontal spacing	4pt
`/csteps/inner color`	color of the text	none
`/csteps/outer color`	color of the circle	none
`/csteps/fill color`	background color of the circle	none

The usage of the package is better explained with examples.

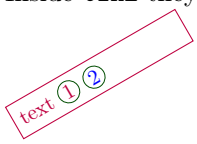
This is normal text: ① is on the baseline, ② is on top.
This is more evident if you have descendents, like p: ⓟ and ⓟ.
You can set options as optional arguments, like Ⓡ or ⓖ.
Math mode is detected too: Ⓐ and Ⓐ^Ⓐ.

```
This is normal text: \Circled{1} is on the
    baseline, \CircledTop{2} is on top.\par
This is more evident if you have descendents,
    like p: \Circled{p} and \CircledTop{p}.\par
You can set options as optional arguments, like \
    Circled[inner color=red]{R} or \CircledTop[
    outer color=green]{g}.\par
Math mode is detected too: \Circled{A} and $\
    Circled{A}^{\Circled{A}}$.
```

For big horizontal things the circle becomes oval: ⑳⓪ or ①⑨⑨.

```
For big horizontal things the circle becomes oval
    : \Circled{200} or \CircledTop{199}.
```

Inside `tikz` they work ok:




```
Inside \texttt{tikz} they work ok:

\tikzset{/csteps/outer color=green!30!black}
\begin{tikzpicture}[scale=0.8, rotate=30,
    text width=3cm, transform shape]
    \node [draw,color=purple]{text \Circled{1}
        \Circled[inner color=blue]{2}};
\end{tikzpicture}
```

If you like more breathing space: ① ② ⓟ ⑳⓪ ①⑨⑨.

```
\pgfkeys{/csteps/inner ysep=10pt}
\pgfkeys{/csteps/inner xsep=10pt}
If you like more breathing space:
\Circled{1} \Circled{2} \Circled{p} \Circled{200}
    \Circled{199}.
\par\bigskip
```

If you want to change colors it's easy: ① ② ⓟ ⑳⓪ ①⑨⑨.

```
\pgfkeys{/csteps/inner color=green}
\pgfkeys{/csteps/outer color=gray}
If you want to change colors it's easy:
\Circled{1} \Circled{2} \Circled{p} \Circled{200}
    \Circled{199}.
\par\bigskip
```

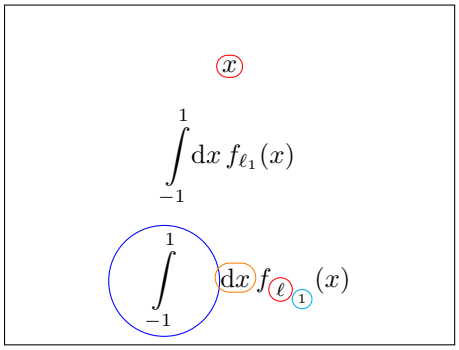
And you can have filled circles and ovals: ❶ ❶❾❾.

```
\pgfkeys{/csteps/inner color=yellow}
\pgfkeys{/csteps/outer color=red}
\pgfkeys{/csteps/fill color=black}
And you can have filled circles and ovals:
\Circled{\textbf{1}} \Circled{\textbf{199}}.
\par\bigskip
```

You can have inline numbers like ① or exponents② and so on. They follow the current color: ① and ② automatically, as you can see: ①+1≈3. In-text circled numbers look better when a bit smaller, though, as you can see in ① for example.

```
You can have inline numbers like \CircledText{1} or
exponents\textsuperscript{\CircledText{2}} and so on.
\textcolor{red}{They follow the current color: \CircledText{1} and \CircledText{2} automatically,} as you can see: \CircledText{$1+1\approx3$}. In-text circled numbers look better when a bit smaller, though, as you can see in {\small\CircledText{1}} for example.
```

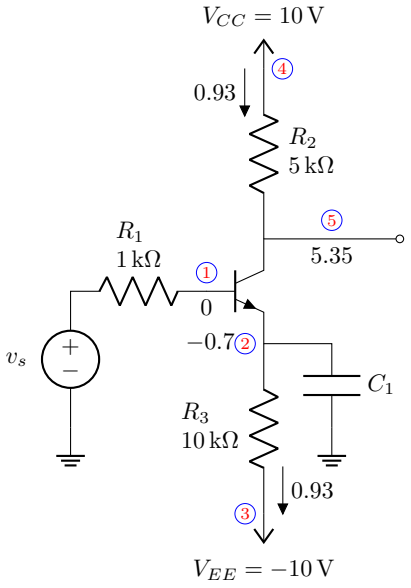
You can have also quite nice math mode effects, thanks to suggestion from `@tallmarmot`:




```
\[\Circled[outer color=red]{x}\]
\[\int\limits_{-1}^1\!\mathrm{d}x\,f_{\ell_1}(x)\]
\[\Circled[outer color=blue]{\int\limits_{-1}^1}\!
   \Circled[outer color=orange]{\mathrm{d}x}
f_{\Circled[outer color=red]{\ell}_{\Circled[outer color=cyan]{1}}}(x)\]
```

3 Automatically generated numbers

The command `\cstep` will generate a circled number, starting from `1`, that can be referenced with the normal `\label`/`\ref` mechanism.[2] You can reset the numbering with `\startcstep`. For example:



①: let's neglect (to be confirmed!) I_B, $V_B \approx 0\,\mathrm{V}$;

③: $I_{E1_Q} = (V_{E1_Q} - V_{EE})/R_3$;

④: $I_{C1_Q} \approx 0.93\,\mathrm{mA}$;

⑤: $V_{CC} - R_2 I_{C2_Q} = 5.35\,\mathrm{V}$; $V_{CE1_Q} = 6.05\,\mathrm{V}$;

[2] Notice that in the example the numbers in the references on the right are blue due to `hyperref` configuration!

```
\tikzset{/csteps/inner color=red, /csteps/outer color=blue}
\newcommand{\circref}[1]{\CircledText{\small\textbf{\ref{#1}}}:}
\begin{circuitikz}[scale=0.9, baseline=(VCC), transform shape]
    \draw (0,0) node[ground](GND){} to[V=$v_s$] ++(0,2)
    to[R, l2^=$R_1$ and \SI{1}{k\ohm}, l2 valign=b] ++(2,0) coordinate(firstb);
    \node [above] at(firstb) {\cstep\label{c:b1}};
    \node [below] at(firstb) {0};
    \draw (firstb) node[npn, anchor=B](Q1){};
    \draw (Q1.E) node[left]{\num{-0.7}\cstep};
    \draw (Q1.C) to[R, l2_=$R_2$ and \SI{5}{k\ohm}, f<=\num{0.93}] ++(0,2.5)
        node[vcc](VCC){$V_{CC}=\SI{10}{V}$} ;
    \draw (Q1.E) to[R, l2_=$R_3$ and \SI{10}{k\ohm},
    f=\num{0.93}] ++(0,-2.5) node[vee](VEE){$V_{EE}=\SI{-10}{V}$}
        node[left]{\cstep\label{c:ie1}};
    \path (VCC) node[right]{\cstep\label{c:ic1}};
    \draw (Q1.E) -- ++(1,0) coordinate(tmp) to[C=$C_1$] (tmp |- GND) node[ground]{};
    \draw (Q1.C) -- ++(1,0) coordinate(g1) to[short, -o] ++(1,0);
    \node [above] at(g1) {\cstep\label{c:g1}};
    \node [below] at(g1) {\num{5.35}};
\end{circuitikz}\quad
\begin{minipage}[t]{0.5\linewidth}
    %% minipage reset these..
    \parindent=0pt\parskip=6pt plus 6pt minus 2pt\relax
    \circref{c:b1} let's neglect (to be confirmed!) $I_B$, $V_B\approx\SI{0}{V}$;\par
    \circref{c:ie1} $I_{E1_Q}=(V_{E1_Q}-V_{EE})/R_3$;\par
    \circref{c:ic1} $I_{C1_Q} \approx\SI{0.93}{mA}$;\par
    \circref{c:g1} $V_{CC}-R_2I_{C2_Q}= \SI{5.35}{V}$; $V_{CE1_Q}=\SI{6.05}{V}$;\par
\end{minipage}
```

4 Personalize it!

The definition of `CircledTexT` is simply the following one; you can get idea and define your own easily (beware of spaces at the end of the lines, though!):

```
\newcommand{\CircledText}[2][]{%
    \CircledParamOpts{inner color=none, outer color=none, #1}{1}{#2}%
}
```

where `\CircledParamOpts` is a macro with three arguments: the first one is the list of options (without the `/csteps/` part), the second one is 1 is you want the circle content on the baseline and 0 if you want it on top of it, and finally the third one is the text to be "circled".

Also the implementation of the `\cstep` command and relatives is quite simple, and you can play a lot with it to change things (formats, colors, the type of numbering, and so on):

```
\newcounter{cstepcnt}
\newcommand{\startcstep}{\setcounter{cstepcnt}{0}}
\newcommand{\resetcstep}{\setcounter{cstepcnt}{0}}
\newcommand{\cstep}{%
    \refstepcounter{cstepcnt}%
    \Circled{\scriptsize\arabic{cstepcnt}}%
}
\renewcommand{\thecstepcnt}{\arabic{cstepcnt}}
```